

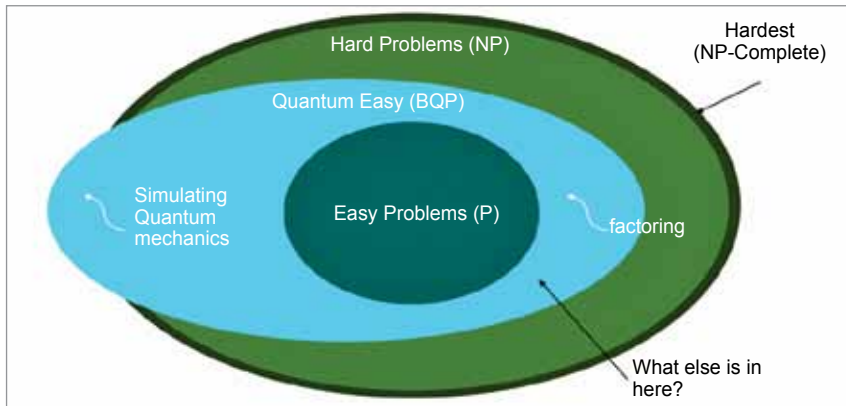


Fig. 4: BQP solvability of quantum computers

rary, classical security primitives, one needs to have a big enough (of many qubit) fault-tolerant and stable quantum computer—which will be happening in near future but not happening tomorrow. Then why should I right now bother about it? The primary reason is, an adversary can potentially steal our today's encrypted messages and can decipher them tomorrow, being powered by quantum oracle. And this sort of development requires a holistic research approach and an overhaul of the cryptographic infrastructure, which are time consuming. So better to start today.

Quantum gadgets

Is the picture only so gloomy? Not really. There is silver lining also. We now have something called Quantum Gadget, which can be used to enhance the security of the classical communication primitives. One such example is Quantum Key Distribution (QKD). The idea of QKD is derived from the fact that two honest parties can have a shared random secret key that is only known to them. If any adversary wants to intercept the secret random key, the adversary must read the key. Once a quantum state is read (measured), it collapses to one of the classical states—either 0 or 1. Hence any such adversary intrusion results in a detectable anomaly in the overall system.

There are other ways also a quantum gadget can pave way into our classical security systems and primitives, such as quantum fingerprints. One can think of it as a technique involving a quantum computer to generate a string like our classical cryptographic

hash functions. There are various other examples also, but not limited to, like quantum random number generation, quantum signature generation, Byzantine agreement, quantum flip-the-coin, secure e-voting, secure multi-party computation, etc.

In nutshell

Quantum computer research, as shown in Fig. 2, is widespread and across nations. One of the areas of immediate interest of these researchers is related to making classical devices, communication, and information to be adaptable and secure in post quantum era. Basically, making them quantum safe.

In this article we started off with discussing how our classical notion of security evolves around RSA. Then we discussed the notion and concept around quantum safety. We discussed the necessary but not sufficient condition of keeping our security primitives outside the reach of quantum solvable BQP domain (Fig. 4). At last, we concluded with an optimism that quantum gadgets like quantum key distribution (QKD) are there to help us in near future to achieve quantum safely.

Of course, the journey has just begun, and it is far from concluded with some definitive answers. **EFY**

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Pradip Mukhopadhyay has 19 years of experience across the stack—from low level systems programming to high level GUIs. He is a FOSS enthusiast, and currently works at NetApp, Bengaluru

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